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# Mental health-aware sentiment analysis using a hybrid quantum–classical approach

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## Abstract

Social media offers a real-time opportunity to examine human emotions, behaviors, and mental health as it becomes more and more integrated into daily life. These platforms are rich sources of public opinions and expressions that provide insightful information about psychological health. Such data have been interpreted using traditional sentiment analysis. However, it frequently fails to identify subtle emotional cues that are important for evaluating mental health. In order to improve sentiment analysis for mental health detection using social media datasets, this study investigates a novel integration of Quantum Natural Language Processing (QNLP). The study proposes a hybrid quantum–classical framework for binary sentiment classification of tweets that incorporates a custom quantum layer into an LSTM network. To enhance the feature representation, the quantum layer employs variational quantum circuits with rotational and entanglement gates. The final classification is achieved by concatenating the results with classical LSTM outputs. In addition, we introduced a recency-aware temporal weighting mechanism to dynamically monitor mental health of users. Our model offers improved feature entanglement capabilities while achieving competitive performance (validation accuracy of 0.7707) when compared to a classical LSTM baseline (0.7775) on the Sentiment140 dataset. This work illustrates the potential of quantum machine learning for scalable, real-time psychological analysis by mapping weighted sentiment scores to inferred mental health states. The suggested framework focuses on large-scale, real-time social media streams, where repeated quantum circuit simulations and training and inference over millions of temporally evolving textual instances require GPU acceleration, parallel execution, and high-performance computing (HPC) resources for practical deployment.

**Keywords** Mental health assessment · Sentiment analysis · Natural language processing

Extended author information available on the last page of the article

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## 1 Introduction

Mental health is a growing problem across globe. It affects populations of all ages and all cultures [1]. Mood disorders (depression, anxiety disorders, and stress-related disorders) are the leading cause of disability worldwide according to the World Health Organization (WHO). This is due to social stigma, lack of awareness, and mental health services. In the worst cases, unrecognized mental illness can lead to self-harm or suicide [2]. These facts lead toward the need for scalable, continuous, and early-stage mental health assessment. Mental health professionals conventionally assess patients using self-reported questionnaires and conduct structured clinical interviews. Though these methods are clinically useful, they have high probability of affected with subjective bias and are expensive. They generally are unable to detect early signs and sudden psychological changes. This passive problem within emotional assessment of individual can be eased using fast-growing social media. The day-by-day data generated by these platforms can help to capture the cognitive state of the users. The real-time data hold the emotional essence of users through thoughts and behavioral patterns. Among these platforms, Twitter allows a person to express their feelings in the form of tweets. These data can be easily assessed to determine the population and individual-level monitoring of personal health [3, 4]. Sentiment analysis is one of the major application of Natural Language Processing (NLP). It is used to figure out the emotional polarity and subjectivity from the raw text. This analysis can infer states like stress, anxiety, and depression directly from users linguistic expression [5, 6]. These cues are still difficult to detect as the language semantics are highly diversified and evolving over time [7]. Today, sentiment analysis is applied across a number of applications in analyzing customer feedback, monitoring social trends, product, and service recommendation systems. This has helped in defining a mean for intelligent interaction between humans and computers [8–11].

### 1.1 Computational challenges in scalable sentiment-based social media analysis

The current deep learning models like RNNs, LSTMs, and transformers have shown strong performance in sentiment classification tasks [12, 13]. Large language models including BERT and GPT variants have pushed contextual understanding further by capturing long-range dependencies and semantic relationships [14–16]. But the performance of these models struggles due to various drawbacks. Their parameter complexity is high and consumes substantial energy. These models also suffer from longer training and inference times. A key issue lies within the inefficient understanding of the obtained results by them, creating a interpretability bottleneck within concerning area. These factors limit the scalability and real-time deployment which is needed in continuous mental health monitoring applications where low latency matters. Such models will face critical barriers during scalable implementation on the voluminous social medial data. Real-time assessment needs continuous ingestion, preprocessing, feature extraction, and inference operations. Temporal aggregation is also needed to track evolving emotional states, which will initiate repeated

recomputation as new data streams in. Single-node hardware's will struggle on this large datasets that exceeds their training and memory capacity. These constraints lead to the requirement of high-performance computing (HPC) infrastructure. The provided parallelism, acceleration, and scalability are essential for seamless integration and assessment of the data in real-time context.

### 1.2 Quantum-enhanced learning paradigm

Recent developments in research are exploring more efficient computing means than classical DL. Among these alternatives, Quantum Computing (QC) has established itself as an reliable approach. It uses quantum properties to represent the information features in high-dimensional space. This allows this technology to execute multiple operations of computation in parallel [17, 18]. QC leads the way for quantum-enhanced semantic and syntactic relationship between words [19]. This quantum processing makes it possible to reduce the trainable parameters by understanding complex correlation more efficiently. However, this approach is still nascent and faces hurdles in acquiring full quantum advantage. The limited count of feasible qubits, their noise susceptibility, and computational necessity to use lower depth circuits in current NISQ era hinder Quantum Natural Language Processing (QNLP) [20]. So, rather than using full quantum approach, current research studies are biased toward using hybrid quantum–classical architectures for more practical application. This hybrid concept has made it possible to run quantum representation within current classical workflows. As of now, there are still very limited research studies that use these approaches for analyzing sentiment within natural language through standard benchmark datasets. This is mainly driven by the computation overhead associated with NISQ hardware. The need for classical simulation of these models makes it essential for the learning pipelines to evaluate the output through multiple runs. This chaotic measurement increases the amount of resources required during training and inference. Even if, these evaluations are well-suited to be evaluated across simultaneous parallel runs. Thus, the hybrid architectures can be modeled through GPU-accelerated and distributed platforms. This way, rather than reducing the direct computational demand, high-performance computing (HPC) can be used to implement these architectures more efficiently. This approach is generally referred to as Quantum-Centric Supercomputing (QCSC). QCSC combined both HPC and quantum computing resources within a single architecture. This technique provides a distributed method for sharing the quantum computation requirements across multiple heterogeneous systems. This is evident in quantum-selected configuration interaction (QSCI) and sample-based quantum diagonalization (SQD). In these techniques, classical HPC is combined together with quantum sampling to solve existing unsolvable problems. These supercomputing platforms are a critical intermediary step between transitioning from NISQ era to future fault-tolerant quantum systems [21]. The current rise in generative artificial intelligence and its foundational models also support the accelerated computing aspect. They use multiple high-performance GPUs for training their native complex architectures like transformer-based GPT [22]. Thus, building on these classical perspectives, the current hybrid QNLP

solutions can be similarly developed through accelerated hardware. This is essential for maintaining true quantum computational advantage in real-time systems.

## 2 Literature review

### 2.1 Classical landscape for mental health-aware sentiment analysis

Social media data had been utilized by a number of studies to assess the current mental health state of an individual. These studies had explored various number of ML architectures like deep learning, CNN, LSTM, transformers, and alike within recent years. As such, Benrouba et al. [23] used a Natural Language Understanding framework to classify textual posts (collected through Twitter API) within five emotional categories (joy, sadness, anger, fear, and disgust). They constructed emotion lexicon (450 terms) to identify and filter emotionally negative content to provide a quality-rich emotional information to the users. Noor et al. [25] had processed social media text to detect the depression state through transformer-based ensemble stack. They combined 6 different pre-trained models, (BERT, RoBERTa, ALBERT, DistilBERT, XLNet, and BART) to process collected English tweets, achieving substantial performance with 96.93% precision, 97.50% recall, 97.22% F1-score, and 97.21% accuracy. Another study by Jacob [29] adopted Bi-LSTM to detect the depression sentiment with superior performance of 98.65% accuracy, 99.13% precision, 98.09% recall, and a 98.6% F1-score on twitter-based data. Likewise, Sidorov et al. [26] developed the first of its kind multilingual multi-class hope speech dataset (MIND-HOPE) that comprises 19,183 Spanish and 21,043 German tweets categorized among four hope categories. The findings establish the need for language-specific fine-tuning in understanding the linguistic characteristics across multilingual domain. They further established the superiority of transformer's as a benchmark models across NLP and sentiment analysis. Fahmi et al. [27] used 10,000 English language tweets to benchmark BERT and RoBERTa models for sentiment classification. The optimized pre-trained latter model outperformed with significant performance (reflected as 90.45% accuracy and 90.39% F1-score,) as compared to its compared counterpart. Notably, Tokac et al. [28] adopted sentiment scoring, aspect-based analysis, and spearman correlation for assessing the relation between mental health and exercise behavior through theme-wise classified (mental health, exercise, and covid-related) twitter data. The findings of study underscored that how individuals utilized physical activity to cope with issues such as depression and crisis. Another study by Odja et al. [24] analyzed pre- and post-pandemic reddit posts through ML classifiers like K-NN, random forest, and neural network to mental health as an sentiment-based assessment. In this study, random forest has best performance (accuracy and F1-score $\approx$ 80.6%), while neural learners had lowest error rate of 0.06496. Howsoever, the current studies lack in scalable generalized models that can effectively assess the multimodal cross-domain and multi-lingual aspects of the problem (Table 1).

### 2.2 Quantum approaches for mental health-aware sentiment analysis

With the advent of quantum natural language processing (QNLP), quantum-enhanced learning models have been used to analyze the semantic context for mental health assessment. Li et al. [30] proposed a multimodal sentiment analysis framework integrating textual and acoustic representations using variational quantum circuits (VQCs) and a residual fusion strategy. Their architecture combined BERT for text encoding and HuBERT for speech representation, followed by transformer-based self-attention and cross-attention mechanisms to align multimodal features. A randomized VQC was employed to process re-uploaded, strongly entangled angle-encoded features on the MELD and IEMOCAP datasets, leading to improved sentiment classification performance. Complementary to multimodal approaches, Sathiya and Arokia Renjit [31] trained Quantum Neural Network (QNN)-based framework to model social media data for predicting mental health states. They combined BERTopic-based stress theme extraction with sentence-BERT-driven personality trait inference and quantum entanglement to capture nonlinear interactions of psychological factors within twitter data. The incorporated authenticity score quantified discrepancies between self-reported and inferred traits, achieving an accuracy of 94.5% in detecting mood disorders such as depression, anxiety, bipolar disorder, OCD, PTSD, and schizophrenia. Another study by Padha et al. [32] benchmarked quantum-LSTM (QSLTM) contrastive learning framework for assessing mental health conditions across seven datasets. Their results demonstrated that QLSTM architectures can effectively capture long-range temporal dependencies through quantum-assisted memory mapping, enabling improved sentiment categorization over time series data.

### 2.3 Motivation for the study

In the above-scoped literature, we found that there are limitations within the present quantum-enhanced architectures. Their scalable performance is complicated due to the need for computationally heavy transformers and complex feature engineering. These models are also trained on small and curated datasets. In addition, these experiments generally report their gains under controlled environments. So, their overall generalized performance during real-time data volume for practical social media monitoring is hindered. Beyond these execution issues, the current NISQ limitations also pose another threat. The current architectures need to consider the circuit depth estimation and design of quantum memory. These architectures efficiency is sensitive to parameters scalability and training during large data. Beyond execution and hardware constraints, a fundamental modeling challenge arises from the nature of complex linguistic feature interactions in sentiment analysis. Such interactions refer to nonlinear, higher-order dependencies among multiple latent linguistic cues whose joint configuration determines sentiment, rather than isolated tokens or purely sequential context. In practical language use, sentiment often emerges from the combined influence of negations, modifiers, discourse framing, and distributed cues that span non-adjacent positions in a sentence (e.g., negation–modifier coupling such as

**Table 1** Related literature associated with machine learning-based sentiment analysis

| References | Data source | Method/model used | Emotion categories | Key contribution | Limitation/gap identified |
|---|---|---|---|---|---|
| Tejaswini et al.[23] | Twitter posts collected via Twitter API | Natural Language Understanding (NLU) API for emotion extraction and classification | Joy, Sadness, Anger, Fear, Disgust | Developed a social media filtering pipeline that detects emotional tone, matches it to a lexicon of 450 emotional words, and identifies potentially harmful content for users | Relies solely on classical NLP tools; limited capability to capture deeper semantics and contextual nuances, especially under real-time constraints |
| Odja et al. [24] | Reddit posts (pre- and post-pandemic) | k-NN, Random Forest, Neural Network | Accuracy, F1-Score, Recall, Precision, Error Rate | Random Forest achieved the best overall performance (F1 = 80.6%, highest accuracy). Neural Network showed lowest error rate (0.06496). k-NN performed worst among the three models | Limited to classical ML techniques; moderate accuracy ceiling; performance may improve with advanced neural architectures or hybrid models and larger, more diverse datasets |
| Noor et al. [25] | English-language social media posts (Twitter/X) | Text | Stacking ensemble of BERT, RoBERTa, ALBERT, DistilBERT, XLNet, and BART (SENSDeep) | Achieved high performance with Precision = 96.93%, Recall = 97.50%, F1-score = 97.22%, Accuracy = 97.21%; outperformed individual transformer baselines | Limited to English text; absence of demographic attributes; generalizability across cultures unverified; future work includes multilingual modeling, demographic-aware analysis, and mobile application deployment |

**Table 1** (continued)

| References | Data source | Method/model used | Emotion categories | Key contribution | Limitation/gap identified |
|---|---|---|---|---|---|
| Sidorov et al. [26] | Twitter (MIND-HOPE dataset) | Spanish, German | Traditional ML (SVM, LR, RF), DL (CNN, BiLSTM), Transformers (BERT, RoBERTa, mBERT, XLM-R) | Monolingual transformers outperform multilingual models; Unigrams + SVM best among ML; Transformers achieve highest F1-scores, especially in binary hope detection; enables fine-grained cross-lingual hope analysis | Multiclass hope detection remains challenging; class imbalance affects performance; multimedia hope expression not explored; ethical and demographic bias considerations require further study |
| Fahmi and Nababan [27] | 10,000 English tweets collected via Twitter API | BERT, RoBERTa | Positive, Negative, Neutral | RoBERTa outperformed BERT across all metrics (Accuracy = 90.45%, F1 = 90.39%); improved handling of informal language due to optimized pretraining | Limited to English-language text; unimodal analysis only; future work suggested for multilingual and multimodal mental health detection |
| Tokac et al. [28] | Twitter posts (pandemic period) | Mental Health, Exercise, COVID-19 | Sentiment scoring, Aspect-based sentiment analysis, Spearman correlation | Early pandemic tweets were COVID-focused; later tweets became more exercise- and mental health-focused; shifting correlation between exercise and mental health sentiment over time; strong lexical association between exercise and depression-related terms | Observational analysis only; causal relationships not established; relies on text-only data; suggests need for public health interventions promoting exercise to mitigate post-pandemic mental health decline |

**Table 1** (continued)

| References | Data source | Method/model used | Emotion categories | Key contribution | Limitation/gap identified |
|---|---|---|---|---|---|
| Jacob [29] | Kaggle: Depression on Twitter | Logistic Regression, SVM, Random Forest, LSTM, Bi-LSTM | Sentiment Analysis + Deep Learning | Bi-LSTM achieved highest performance (Accuracy = 98.65%, Precision = 99.13%, Recall = 98.09%, F1 = 98.6%) | Bi-LSTM outperformed all baselines due to bidirectional context learning; traditional ML models struggled with nuanced emotional patterns |

"not entirely unacceptable" or contextual polarity inversion such as "hardly impressive for a flagship product"). Capturing these dependencies requires representations that go beyond simple additive composition or pairwise temporal correlations. Even if, classical LSTMs and BiLSTMs are effective at modeling sequential dependencies and contextual flow, but their gating mechanisms primarily operate through additive and element-wise transformations within a fixed latent space. This limits the expressivity in scenarios involving sentiment depending on correlated feature interactions across multiple latent dimensions. While deeper or wider recurrent models may partially mitigate this limitation, they do so at the cost of increased parameterization and reduced training stability, particularly under large-scale data regimes. To address these challenges, we propose a lightweight hybrid quantum–classical architecture that augments a classical sequential encoder with a shallow variational quantum circuit. The quantum module is designed to act as a nonlinear feature transformation layer, introducing correlated and entangled interactions among compressed semantic features without replacing classical sequence modeling. Our architecture used constrained circuit depth and qubit count to maintain computational feasibility under NISQ-era simulation constraints while selectively enhancing representational expressivity. The resulting hybrid representations are subsequently passed to a temporal aggregation module, enabling longitudinal analysis of user sentiment dynamics for real-time and large-scale social media monitoring.

## 3 Learning models

### 3.1 Long short-term memory (LSTM)

The LSTM is designed to overcome the vanishing gradient issue present in the traditional recurrent neural network (RNN). Because standard RNNs cannot sustain informative gradients over long temporal spans, they become unstable when learning long-range dependencies. LSTMs help to solve this issue by introducing an explicit cell state that enables stable gradient propagation and a system of multiplicative gates that regulate information flow, as illustrated in Fig. 1. This cell state mechanism enables these temporal learners to have a choice between retaining, updating, or discarding a particular instance of information over time. This property makes them a viable choice for task that requires sequential modeling such as sentiment analysis and health-related temporal correlations.

At each time step $h_{t-1}$, the LSTM receives the current input in the form of $x_t$, the previous cell state $c_{t-1}$, and hidden state $h_{t-1}$. Particularly, the LSTM module comprises a forget gate, input gate, candidate cell update, and output gate operations. They work together to determine the hidden representation generated at that time step as well as the internal memory update.

*a) Forget gate (Managing Memory Retention)* This gate helps the model in deciding that which of the previous cell states need to conserved over time. It calculates a gating vector $f_t \in (0, 1)^d$, where values near 0 suppress prior information and values near 1 preserve it:

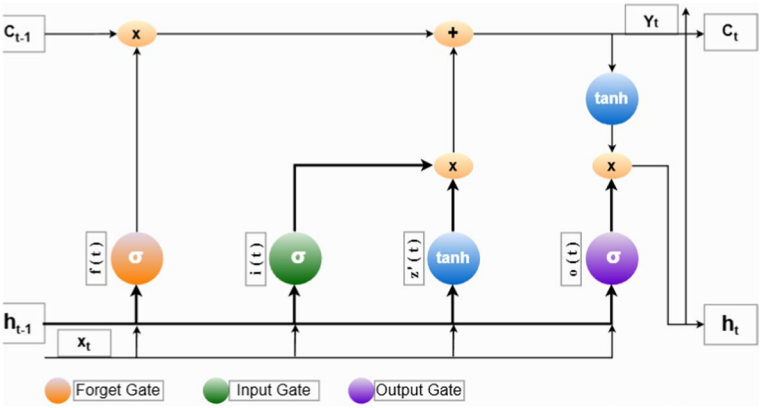



**Fig. 1** Computational architecture of LSTM

$$f_t = \sigma\big(W_f[h_{t-1}, x_t] + b_f\big), \tag{1}$$

where $W_f$ and $b_f$ are the learned weights and bias, and $\sigma(\cdot)$ represents the logistic sigmoid function. The forget gate enables the network to selectively preserve long-term contextual cues while eliminating unnecessary or out-of-date information by performing an element-wise multiplication with $c_{t-1}$

*b) Input gate and candidate cell state (Incorporating New Information)* The input gate, which is made up of two parts, controls the addition of new data to the cell state. The activation gate for input is provided by:

$$i_t = \sigma\big(W_i[h_{t-1}, x_t] + b_i\big), \tag{2}$$

which establishes the amount of new information added. The proposed candidate cell is calculated as follows:

$$\tilde{c}_t = \tanh\big(W_c[h_{t-1}, x_t] + b_c\big), \tag{3}$$

produces a set of candidate features that could be added to the cell state. Numerical stability is maintained by the hyperbolic tangent activation, which guarantees that the candidate values fall within the interval $[-1, 1]$.

*c)Cell state update ( Memory Evolution Across Time)* To update the internal memory of the LSTM itself, the retained past information is combined together with the newly selected information, as follows,

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{4}$$

where $\odot$ denotes element-wise multiplication. This mathematical construct helps the LSTM to learn long-range dependencies more efficiently. This multiplicative gating structure restricts the gradients from vanishing during backpropagation across time. This allows this model to preserve semantic dependencies over multiple steps.

*d) Output gate and hidden state ( Producing the Time-Step Representation)* The output gate determines which parts of the updated cell state are exposed to the rest of the network:

$$o_t = \sigma\big(W_o[h_{t-1}, x_t] + b_o\big). \tag{5}$$

The output of LSTM at a time step $t$ is described by the hidden state $h_t$ as follows:

$$h_t = o_t \odot \tanh(c_t). \tag{6}$$

In contrast to the cell state $c_t$ which encodes long-term memory, the hidden state $h_t$ provides a contextual representation as per the immediate prediction task, such as sentiment classification at the tweet level.

### 3.2 Bidirectional long short-term memory (BiLSTM)

While a standard LSTM processes input sequences from past to future, it is inherently limited by its unidirectional nature: the hidden representation at time step $t$ depends only on information available in earlier time steps. However, a lot of linguistic patterns depend on both preceding and succeeding context. This issue is resolved by adding a parallel LSTM layer to process the sequences in reverse order simultaneously. This integration of forward and backward data pass help in developing enriched contextual embedding within BiLSTM module.

The foundational architecture of BiLSTM is displayed using Fig. 2. It contains two parallelly running LSTM instances, where forward LSTM chain computes hidden state sequence using

$$\overrightarrow{h_t} = \mathrm{LSTM}_{\mathrm{fwd}}(x_t, \overrightarrow{h_{t-1}}, \overrightarrow{c_{t-1}}),$$

while the backward LSTM processes the sequence in reverse:

$$\overleftarrow{h_t} = \mathrm{LSTM}_{\mathrm{bwd}}(x_t, \overleftarrow{h_{t+1}}, \overleftarrow{c_{t+1}}).$$

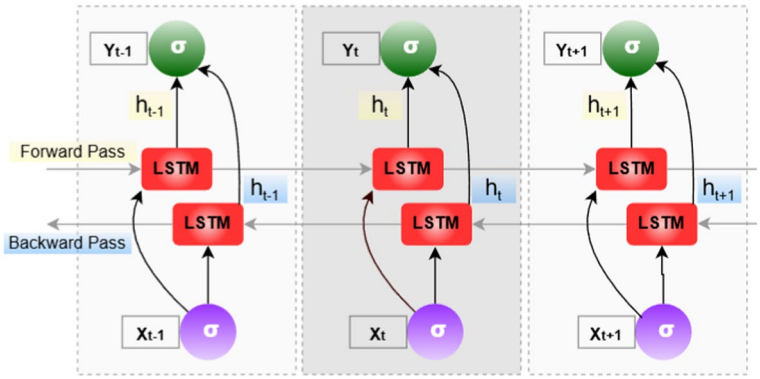



**Fig. 2** Computational architecture of bidirectional LSTM

The final contextual representation at time $t$ is formed by concatenation:

$$h_t = [\overrightarrow{h_t} \ \overleftarrow{h_t}],$$

which encodes both historic and future dependencies relevant to the token $x_t$.

*a) Internal LSTM operations within BiLSTM.* Each directional LSTM follows the gating operations described in the previous subsection. For completeness, the internal transformations are summarized here for an input $x_t$, previous hidden state $h_{t-1}$, and previous cell state $c_{t-1}$:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \tag{7}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \tag{8}$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c), \tag{9}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{10}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \tag{11}$$

$$h_t = o_t \odot \tanh(c_t). \tag{12}$$

These equations describe the overall computation construct of both the forward and backward subnetworks. The forward LSTM learns relationships based on prior context, while the backward LSTM learns patterns dependent on subsequent context. Their combined representation significantly strengthens modeling capability in tasks such as sentiment analysis, where clues may appear before or after a key token. Due to architectural advantage of BiLSTM over LSTM, it is useful in tasks involving natural language processing, like sentiment analysis. It achieves contextual symmetry by processing sequences in both forward and backward directions. This enables BiLSTM to capture the inter-dependencies within data that a unidirectional LSTM fails to understand. This bidirectional processing improves the semantic resolution, especially for linguistic constructs whose interpretation depends on future context, such as intensifying expressions and negations. The combination of two independent recurrent pathways increases robustness and further reduces the model's sensitivity to noisy or poorly informative tokens. In the proposed hybrid architecture, the BiLSTM functions as the primary contextual encoder, generating a rich classical representation that is subsequently improved by the quantum feature enhancement module.

### 3.3 Quantum-based variational circuits design

Quantum computing uses a quantum mechanic-based representation to perform computations in comparison to classical neural networks. This approach used

operations in complex-valued Hilbert spaces that are governed by unitary transformations. A quantum bit (qubit) is described by the superposition state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\gamma|^2 + |\delta|^2 = 1, \tag{13}$$

where the probability amplitudes $\gamma$ and $\delta$ encode information beyond classical binary representations. These qubits can be entangled together using quantum gates such as Hadamard gate in to non-factorizable quantum state. This allows the associated computation operation to capture non-classical correlation. This way, it enables a compact modeling of high-order feature interactions.

In the proposed hybrid architecture, a custom-featured-based variational quantum circuit is embedded within a BiLSTM-based text classification pipeline. Rather than operating directly on raw input tokens, the quantum circuit receives a compressed latent representation obtained after classical sequence modeling. This design allows the quantum layer to focus on transforming semantically meaningful features while maintaining computational feasibility.

#### 3.3.1 Parametric rotation encoding

$$\mathbf{z} = (z_1, z_2, \ldots, z_n) \tag{14}$$

denote the reduced feature vector obtained after global temporal pooling of the BiLSTM output. These $z_i$ components are then normalized and embedded into the quantum state via a single-qubit rotation around the $y$-axis:

$$R_y(z_i) = \begin{pmatrix} \cos\left(\frac{z_i}{2}\right) & -\sin\left(\frac{z_i}{2}\right) \\ \sin\left(\frac{z_i}{2}\right) & \cos\left(\frac{z_i}{2}\right) \end{pmatrix}. \tag{15}$$

By applying the angle encoding, the real-valued classical data values are then transformed into quantum-based constructs. These constructs are relatively aligned with the latent semantic structures learned by the classical encoder.

#### 3.3.2 Custom entanglement features

To introduce correlations between encoded features, the circuit applies a linear chain of controlled-NOT (CNOT) gates, entangling adjacent qubits:

$$\mathrm{CNOT}_{i,i+1}, \qquad i = 1, \ldots, n-1. \tag{16}$$

This entanglement pattern has been introduced using a custom feature map using angle rotation and CNOT gate for variational, trainable quantum embeddings rather than fixed kernel constructions. The resulting variational circuit enables the model to learn task-adaptive correlations between compressed semantic features extracted from the classical encoder. This quantum entanglement induces non-separable feature interactions, such that the expectation value measured on a single qubit depends on the joint configuration of multiple encoded features. In contrast to classical linear or independently activated transformations, entangled quantum states encode global

correlations across feature dimensions. These quantum interactions help in capturing interdependent semantic cues, such as mixed sentiment expressions, contextual polarity shifts, or linguistic ambiguity, where sentiment cannot be attributed to a single token or local context alone. The quantum entanglement layer operates on a temporally pooled representation and adds cross-feature coupling that is invariant to token order, in contrast to classical LSTM hidden states that mainly model sequential dependencies through gated recurrence. Thus, during scenarios involving when positive and negative sentiment indicators coexist within the same utterance, this enables the model to encode higher-order interactions between semantic components that may be missed by purely sequential processing.

#### 3.3.3 Measurement and hybrid feature fusion

Expectation values are obtained by measuring each qubit in the Pauli-$Z$ basis after unitary evolution.

$$q_i = \langle \psi | Z_i | \psi \rangle, \qquad i = 1, \ldots, n. \tag{17}$$

These measurements represent a nonlinear transformation of the initial BiLSTM embedding caused by quantum superposition and entanglement, mapping the quantum state back into a real-valued feature vector. A hybrid feature space is created by concatenating the resulting quantum feature vector with the classical pooled representation. Fully connected layers then process this feature space for classification. Importantly, the parameters of both the classical network and the quantum circuit are optimized jointly via gradient-based backpropagation using PennyLane's differentiable quantum node interface, enabling end-to-end training of the hybrid architecture. The choice of angle-based encoding in the proposed quantum layer is a principled design decision rather than an ad hoc selection. Because of its simplicity, generalizability, and robustness across heterogeneous data modalities, parametric rotation (angle) encoding is one of the most popular and successful methods for mapping classical data into quantum Hilbert space, according to recent surveys and empirical studies in quantum machine learning. By using single-qubit rotation gates to directly embed real-valued features, angle encoding produces shallow circuits that are ideal for simulation and optimization given the limitations of current hardware. Due to the practical limitations of NISQ-era devices, the majority of modern quantum learning frameworks rely on shallow variational circuits made up of rotation and entangling gates and are assessed using quantum simulators like PennyLane or Qiskit [33]. We have used $R_y$-based angle encoding and lightweight entanglement considering the similar empirical studies. This allowed us to ensure that our approach is numerically stable with adequate expressive capacity. We have further intentionally confined the custom variational circuit to a five-qubit configuration with shallow depth. This architectural choice ensures stable optimization, controllable training time, and compatibility with large-scale simulation on conventional high-performance computing resources. While more expressive circuits with deeper entanglement patterns, more rotational layers, or more qubits might increase representational power, these extensions would worsen training instability and cause a large computational overhead.

This principle configuration establishes a scalable foundation for future exploration of deeper and more expressive quantum circuits on supercomputing platforms.

## 4 Methodology

### 4.1 Problem formulation and framework

- For sentiment analysis of tweets, develop a hybrid quantum–classical model that combines a custom quantum circuit with an LSTM.
- To ensure that recent tweets are relevant for dynamic mental health assessment, prioritize them using a temporal weighting system.
- Create an end-to-end pipeline that links weighted sentiment scores to positive, neutral, and negative mental health states.
- Provide a reusable implementation for QML-NLP applications that are more widely used.

### 4.2 Proposed framework

Figure 3 illustrates the proof-of-concept hybrid quantum–classical framework developed for tweet-level sentiment classification and downstream mental health state estimation. The system integrates conventional LSTM-based sequence modeling with a lightweight quantum feature enhancement module and a temporal reasoning block, resulting in a recency-aware assessment pipeline that captures both linguistic sentiment and temporal patterns in user's social media activity. The suggested framework takes advantage of accelerated and parallel computing in following manner. Tokenization and text preprocessing are done in batch-parallel. While quantum circuit evaluations are carried out independently for each sample, allowing for data-level parallelism, model training makes use of GPU-accelerated LSTM computation. The requirement for high-performance and distributed execution environments is further reinforced by the need for multiple experimental runs for statistical significance. This establishes the ideality of the current framework to be easily adopted across on cloud-based HPC clusters or supercomputing platforms.

#### 4.2.1 Data preparation

We have used curated subset of the publicly available Sentiment140 corpus, which originally comprises approximately 1.6 million tweets for sentiment polarity. To enable the development of model under controlled environment setting, we constructed a balanced subset of 500,000 English-language tweets, uniformly sampled across sentiment classes. To ensure equal class representation and mitigate the effects of class imbalance, 250,000 tweets labeled as negative sentiment (encoded as "0") and 250,000 tweets labeled as positive sentiment (originally annotated as "4" and remapped to "1") are chosen. This dataset subset was split into training, validation, and test splits using a 60/20/20 ratio that produced 300,000 training samples,

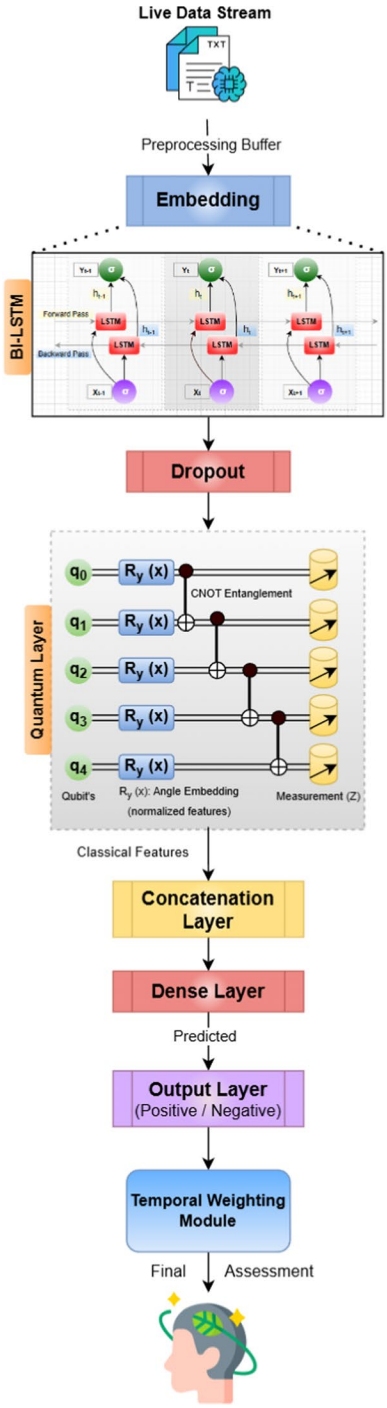



**Fig. 3** Proposed hybrid quantum–classical framework

100,000 validation samples, and 100,000 test samples. Out of these, 250,000 tweets labeled as positive sentiment (originally annotated as "4" and remapped to "1") and 250,000 tweets as negative sentiment (encoded as "0") are selected to ensure equal class representation and mitigate the effects of class imbalance. Stratified sampling guarantees that each split preserves class balance with 50,000 samples per class in the validation and test sets and 150,000 negative and 150,000 positive samples in the training set. This explicit partitioning not only enables early stopping during training, but it also enables reliable performance evaluation. We have used a tokenizer with a fixed vocabulary size of 10,000 to perform text preprocessing. Each of the tweet is then transformed into a sequence of index values and padded or truncated to a maximum length of 50 tokens. This representation is essential for minibatch training as well as compatibility with the bidirectional LSTM encoder and quantum feature transformation module. Then min-max normalized temporal difference between tweets is transformed into exponential decay weights that function as recency coefficients during longitudinal aggregation. This recency-aware weighting scheme prioritizes recent expressions while gradually attenuating older signals, enabling the creation of smooth and understandable user-level sentiment trajectories. The resulting dataset supports rigorous training and evaluation of both classical and hybrid quantum–classical models under consistent experimental conditions. It should be noted here that Sentiment140 is a general-purpose sentiment dataset and does not contain clinically validated mental health annotations. The binary labels lack due to granular emotional details, and thus, inferred sentiment trajectories should be interpreted as proxies for affective expression rather than as diagnostic indicators of mental health.

#### 4.2.2 Hybrid model architecture

The proposed hybrid architecture follows a quantum–classical design in which a deep sequential neural network performs contextual feature extraction, while a lightweight variational quantum module augments the resulting representation with additional nonlinear transformations. Rather than replacing the recurrent sequence model or introducing quantum recurrence, the quantum component operates on a compressed semantic embedding produced by the classical encoder. This design decision guarantees compatibility with NISQ-era constraints, training stability, and scalability. Each input token is projected into a 128-dimensional continuous vector space by the model's first embedding layer. A 128-unit bidirectional LSTM processes the embedded sequence, allowing contextual dependencies to be extracted from both preceding and succeeding tokens. Global average pooling is used to aggregate the sequence of BiLSTM outputs across the temporal dimension in order to produce a fixed-dimensional semantic representation appropriate for large-scale training. To reduce overfitting, a dropout rate of 0.5 is used. A fully connected layer is then used to project the pooled classical representation into a low-dimensional latent space, creating a compact feature vector that is compatible with quantum processing. This reduced representation parameterizes a variational quantum circuit based on the custom feature map, implemented using a five-qubit register. Classical features are embedded into the quantum state via parametric single-qubit rotations, followed by a shallow linear entanglement

structure. Expectation values measured in the Pauli-$Z$ basis yield a quantum-transformed feature vector that captures nonlinear interactions induced by superposition and entanglement, as detailed in Sect. 3.3. The resulting quantum feature vector is concatenated with the pooled classical BiLSTM embedding to form a unified hybrid representation. This fused feature space is processed by a fully connected layer with 64 ReLU-activated units, followed by a softmax output layer that produces sentiment class probabilities. All classical and quantum parameters are optimized jointly in an end-to-end manner using the Adam optimizer with an initial learning rate of $1 \times 10^{-4}$ and sparse categorical cross-entropy loss.

#### 4.2.3 Temporal aggregation and behavioral metrics

We integrated a temporally aware aggregation method along with our hybrid quantum-enhanced classifier. This way, rather than providing a instance-based sentiment prediction, our approach can capture longitudinal behavioral patterns in users through social media.

*a) Recency-weighted sentiment aggregation* Given a sequence of tweets posted by a user, temporal relevance is modeled through an exponential decay function that assigns higher importance to recent posts. Let $d_i$ be the timestamp of the $i$-th tweet and $\Delta_i$ be the normalized time difference between $d_i$ and the reference time. The calculation of temporal weights is as follows:

$$w_i = \exp(-\alpha \cdot \Delta_i),$$

where the decay rate is controlled by alpha. This way, we can ensure that significant impact of the older tweets gradually disappears over the current state for a true real-time assessment. Let $v_i \in [-1, 1]$ represent the predicted continuous sentiment score for tweet $i$. At time $t$, the recency-weighted sentiment trajectory $S_t$ is calculated as follows:

$$S_t = \frac{\sum_{i \le t} w_i \cdot v_i}{\sum_{i \le t} w_i}.$$

Based on this calculation, we produced a smoothened temporal sentiment trajectory to show shifting behavioral pattern rather that expressive noise

*b) Emotional volatility estimation* The rolling standard deviation of sentiment scores over a predetermined temporal window is used to measure emotional volatility. Volatility at time $t$ is defined as follows for a window of size $k$

$$V_t = \sqrt{\frac{1}{k} \sum_{j=t-k+1}^{t} (v_j - \bar{v})^2},$$

where $\bar{v}$ denotes the mean sentiment within the window. Regardless of the direction of sentiment, this metric measures how consistently or inconsistently emotions are expressed.

*c) Trend and drift estimation* The recency-weighted sentiment trajectory $S_t$ is fitted with a linear regression to describe long-term sentiment evolution. The observed positive and negative values tell rising and falling sentiment in semantic manner. The overall slope traversed by these transitions helps to identify the drift change among users.

*d) Behavioral regime definition* Users are divided into behavioral regimes based on the combination of average emotional volatility and sentiment drift direction. The regime definitions used in the analysis are compiled in Table 2. This taxonomy offers an organized and repeatable framework for analyzing temporal dynamics at the user level.

### 4.3 Mental health prediction

Following sentiment prediction, the system computes a weighted sentiment score for each user by multiplying tweet-level sentiment values with their corresponding temporal decay weights and normalizing by the total weight. This score reflects a temporally adjusted sentiment trend and is mapped onto a coarse-grained mental health state classification. Specifically, strongly positive scores are interpreted as indicators of favorable emotional expression, while strongly negative scores suggest the opposite, and intermediate values are assigned to a neutral category. We acknowledge that the proposed sentiment polarity is not equivalent to clinical diagnosis. But, it should be interpretive in terms of understanding affective tendencies of a particular user. Thus, the method is a proof of concept for the potential contribution of quantum-enhanced NLP models to scalable psychological monitoring systems, especially when combined with longer-term digital traces or richer datasets. Clinically annotated data will be incorporated into future developments to enhance interpretability and reliability in actual mental health settings.

### 4.4 Training and evaluation

Using a batch size of 128, we trained our model for a maximum of 20 epochs with early stopping in order to balance computational efficiency and convergence stability. For minibatch training, all input sequences are either padded or truncated to a maximum length of 50 tokens to guarantee uniform dimensionality. To ensure balanced class distributions across all subsets, the dataset is divided using a stratified split into 60% training, 20% validation, and 20% test sets. All models are optimized using the Adam optimizer, which begins with a learning rate of $1 \times 10^{-4}$. A ReduceLROnPlateau scheduler is used to stabilize convergence by reducing the learning rate by a factor of 0.5, with a minimum learning rate of $1 \times 10^{-6}$, when the validation loss does not improve for two consecutive epochs. Based on validation loss, early stopping with a patience of three epochs is used to restore the best-performing model weights while avoiding overfitting and needless computation. To ensure a fair and controlled comparison, the proposed hybrid model and the classical BiLSTM baseline are trained under identical optimization and data-splitting (train-test-val) settings. Five separate runs of each experiment are conducted. While distinct random initializations are applied across runs to capture run-to-run variability, a fixed global random seed is

**Table 2** User behavioral regimes based on temporal sentiment and emotional volatility

| Temporal sentiment trend | Emotional volatility | Behavioral interpretation |
|---|---|---|
| Stable (Near-Zero Drift) | Low | Consistent affective expression with minimal temporal variation, indicative of emotionally stable behavior over time |
| Stable (Near-Zero Drift) | High | Emotionally expressive or reactive behavior without sustained directional sentiment change; short-term fluctuations dominate |
| Directional Drift (Positive or Negative) | Low | Gradual and stable emotional evolution, suggesting consistent sentiment shifts rather than episodic variability |
| Directional Drift (Positive or Negative) | High | Volatile emotional expression coupled with long-term sentiment change, reflecting dynamic and unstable affective behavior |

used to guarantee reproducibility. The reported results are expressed using the mean ± standard deviation for each of these runs. Only the held-out test set is used to evaluate common classification metrics like accuracy, precision, recall, and F1-score. We have evaluated the training curves and confusion matrices to grasp a more detailed understanding of error distribution and learning pattern. This overall study has been carried out on Kaggle environment for computationally management performance.

**Algorithm 1** Hybrid BiLSTM + Quantum Feature-Extraction Framework

**Require:** Dataset $D = \{(t_i, y_i, d_i)\}_{i=1}^{N}$; vocabulary size `VOCAB_SIZE`; maximum sequence length `MAX_WORD_LENGTH`; number of qubits $N_{\text{QUBITS}}$; training hyperparameters; temporal decay factor $\alpha$.
**Ensure:** Trained model $\mathcal{M}$; tweet-level predictions; weighted sentiment score $S$; mental-health label.

**Preprocessing**
Load dataset $D$ and extract tweet text, sentiment labels, and timestamps.
Map sentiment labels: $4 \rightarrow 1$, keep $0$ unchanged.
Fit tokenizer (vocabulary size = `VOCAB_SIZE`) on training text.
Convert tweets to integer sequences and pad or truncate to `MAX_WORD_LENGTH`.
**Classical Feature Encoding**
Embed input sequences using an embedding layer.
Encode sequences using BiLSTM(128) with bidirectional context modeling.
Apply Dropout(0.5) for regularization.
Compress sequence representation using LSTM(64) to obtain latent vector $h$.
**Quantum Feature-Extraction Module (CF2)**
*(Applied once per sample on pooled latent features; not recurrent across time)*
Normalize latent vector $h$ to obtain $h'$.
**for** each qubit $i = 1$ to $N_{\text{QUBITS}}$ **do**
    Encode feature $h'_i$ using parameterized $RY(h'_i)$ rotation.
**end for**
Apply linear CNOT entanglement across qubits.
Measure Pauli-$Z$ expectation values to obtain quantum feature vector $q$.
**Hybrid Feature Fusion and Classification**
Concatenate classical features $h$ and quantum features $q$.
Apply Dense(64, ReLU) layer.
Apply Dense(2, softmax) layer to obtain class probabilities.
Compile model $\mathcal{M}$ using Adam optimizer and sparse categorical cross-entropy loss.
**Training**
Split dataset into training, validation, and test sets.
Train $\mathcal{M}$ with early stopping and learning-rate scheduling.
Save trained model and tokenizer.
**Tweet-Level Sentiment Prediction**
**for** each tweet $t_j$ **do**
    Convert $t_j$ to padded sequence $x_j$.
    Compute prediction $p_j = \mathcal{M}(x_j)$.
    Assign sentiment label $\hat{y}_j = \arg\max(p_j)$.
**end for**
Evaluate accuracy, precision, recall, F1-score, and confusion matrix.
**Temporal Weighting and Mental-Health Assessment**
Map predicted sentiment labels to values $v_j \in \{-1, +1\}$.
Compute recency $\Delta t_j = \text{now} - d_j$.
Normalize recency values using Min–Max scaling.
Compute temporal weights: $w_j = \exp(-\alpha \Delta t_j)$.
Compute weighted sentiment score:

$$S = \frac{\sum_j w_j v_j}{\sum_j w_j}.$$

Assign mental-health label:

$$\text{Positive if } S > 0.5; \quad \text{Negative if } S < -0.5; \quad \text{Neutral otherwise.}$$

**return** $\mathcal{M}$, $S$, and mental-health label.

## 5 Performance metrics

Model performance is evaluated using following standard classification metrics commonly adopted in scholarly studies.

1) *Accuracy*: Accuracy is used to measure the proportion of clearly predicted labels of the sample (e.g., test set) to that of the original true labels. It helps in quantifying the overall performance of the model.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

2) *Loss* Model optimization is guided using sparse categorical cross-entropy loss, which quantifies the discrepancy between predicted class probabilities and the ground truth labels:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right]$$

3) *Confusion matrix* The confusion matrix supports in understanding the true versus predicted class labels structure (False Positives: FP; False Negatives: FN; True Positives: TP; True Negatives: TN), thereby highlighting the actual class-wise prediction patterns and misclassification instances by the model.
4) *Precision, Recall, and F1-score* Aside from accuracy, precision, recall, and F1-score are used scientifically to evaluate the true performance of the model. Precision details out the fraction of true positive predictions over the full predicted positives. Recall, generally understood as sensitivity, measures the effectiveness of the model to detect all of the appropriate positive instances. The F1-score provides the performance intuition through calculating the harmonic mean of precision and recall together. These three metrics can be calculated using the formulations described below:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

5) *Visualization* For clearer understanding, the training and validation accuracy, loss curves, respective confusion matrices are visualized to examine their convergence behavior, stability, and class-wise prediction patterns.

## 6 Results

### 6.1 Classical versus quantum-enhanced model performance

Table 3 lists the performance metrics for our proposed architecture over the Sentiment140 dataset. These values are equal to the mean calculated across 5 independent run for these models along with their standard deviation. Our baseline classical model reached the score of $0.7986 \pm 0.0020$ (accuracy) and $0.7985 \pm 0.0024$ (F1-score). Simultaneously, our proposed hybrid quantum-enhanced model performed in comparison with the classical baseline with an approx. similar values of $0.7987 \pm 0.0016$(accuracy) and $0.7986 \pm 0.0051$ (F1-score). These models also achieved an consistent precision and recall value across independent runs. This signifies that they performed comparably across positive and negative sentiment classification. These obtained result reflects that the quantum enhancement did not lead to absolute improvement in the classical architecture. In contrast, our model establishes that replacing heavy classical models with more efficient and shallow quantum architectures does not reduce the discriminative capability of the classical learners. If we consider the training curves presented in Fig. 5, it is evident that our model achieved stable convergence over the training period. Initially, both of the models displayed a steady increase in their validation accuracy. As the models reached the convergence stage, these models get well saturated. The number of epochs used by the quantum-enhanced learner is less than classical baseline. This suggests that it converged faster as compared to its counterpart. This observed trend is further replicated in loss dynamics during validation stage which is displayed in Fig. 5b. During early epochs, the validation loss of both models decreased rapidly before stabilizing gradually. It is important to note that quantum-enhanced model does not show any unstable loss behavior; thereby, such model's can be easily considered for large-scale recurrent training architectures. For understanding a true classification performance of quantum-enhanced model, its confusion matrix should be properly evaluated. Figure 4 displays the classification instance of this model on the held-out-test dataset. Out of the total samples, the model classified 38,945 negative and 40,889 positive samples correctly. In contrast, it classified 11,055 samples as false negatives and 9,111 samples as false positives. Considering these classification instances along with achieved precision and recall values, it can be noted that model is performing in well-balance manner without any class bias. Although there are lot of false classification instances, it should be noted that the used quantum classifier used merely 5 qubits with simple entanglement. Thus in future work, exploring more number of qubits deeper entanglement structure will provide a substantial performance boost, though this majorly depends upon efficiency of NISQ capabilities.

**Table 3** Performance of classical and quantum models (mean ± standard deviation)

| Model | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| Classical model | $0.7986 \pm 0.0020$ | $0.7994 \pm 0.0132$ | $0.7979 \pm 0.0174$ | $0.7985 \pm 0.0024$ |
| Quantum model | $0.7987 \pm 0.0016$ | $0.7995 \pm 0.0186$ | $0.7987 \pm 0.0283$ | $0.7986 \pm 0.0051$ |

### 6.2 Feature representation

We have analyzed the latent feature representations learned by the classical and hybrid quantum–classical models. Specifically, we extract intermediate embeddings from the pooled BiLSTM layer in the classical pipeline and from the concatenated BiLSTM–quantum representation in the hybrid pipeline, enabling a controlled comparison of feature spaces without retraining or architectural changes. Figure 6 illustrates two-dimensional t-SNE projections of these embeddings using a balanced subset of 3,000 samples from the test data. The representations show structured organization corresponding to sentiment polarity, as predicted by the previously mentioned results, suggesting that discriminative information is maintained across pipelines. The nonlinear transformation caused by the variational quantum circuit is reflected in the hybrid quantum–classical representation's unique geometric arrangement with respect to the classical embedding. We use the silhouette score to measure feature separability in order to supplement the qualitative visualization. The classical BiLSTM features achieve a silhouette score of 0.2870, while the hybrid quantum–classical features achieve a comparable score of 0.2817. Although no improvement in class separability is observed, these results demonstrate that the quantum-enhanced feature transformation preserves latent class structure while inducing an alternative embedding geometry.

These findings support the interpretation of the quantum layer as a complementary feature transformation mechanism rather than a direct performance-enhancing component. The quantum module introduces correlated, non-separable interactions among compressed semantic features without degrading representation quality, aligning with the broader objective of exploring architectural feasibility and representational diversity under hybrid quantum–classical learning paradigms.

### 6.3 Computational complexity and runtime analysis

Both of the models explored within our study were evaluated under similar experimental conditions. They have been trained on identical dataset size with a shared preprocessing pipeline. The used batch size, optimizer configuration, and the early stopping criteria are also kept fixed so that we can ensure a fair and controlled comparison among them.

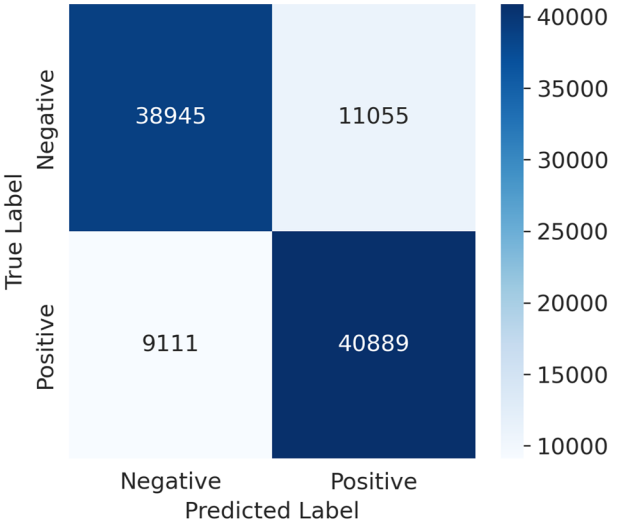
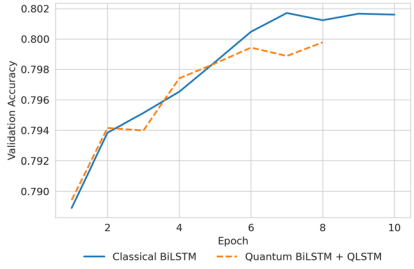
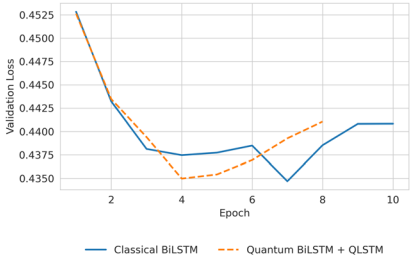



**Fig. 4** Confusion matrix for the quantum model evaluated on the held-out test dataset




(a) Validation accuracy comparison between classical and quantum models.




(b) Validation loss comparison for classical and quantum models during training.

**Fig. 5** Comparison of training dynamics for classical and quantum models. **a** Validation accuracy. **b** Validation loss

### 6.3.1 Empirical runtime analysis

These models are trained and validated across 5 independent runs on a subsampled dataset of 500,000 tweets. The classical model in average required 7.8 training epochs that run for over $5044 \pm 1150$ seconds per training run approximately. Considering the observed per-minibatch execution time, that is nearly equal to 275 ms, it can be attributed that embedding lookup and BILSTM inherent operation dominated the recurrent sequence modeling. The quantum-enhanced model in comparison needed slightly more training time per run with an average of $7020 \pm 820$ seconds. In a similar contrast, these models also required near 6–7 epochs for stabilized convergence in each run and 495 ms of per-minibatch execution time. This observed increase in training time of 39% is due to overhead presented by the simulation of the gate-level circuits

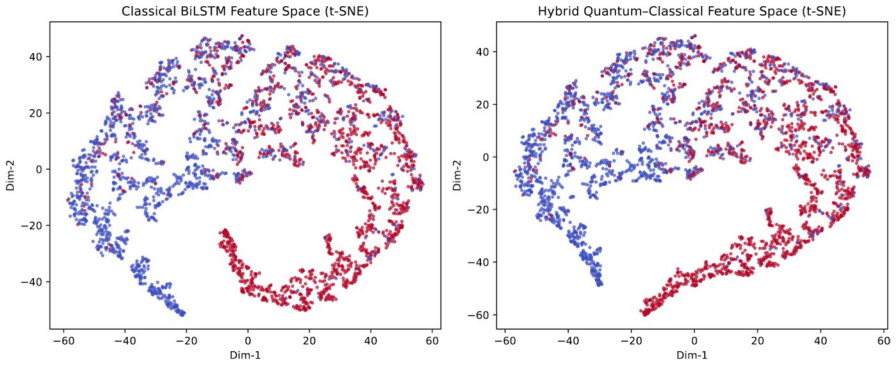



**Fig. 6** Feature space visualization across proposed architecture

of quantum classifier. This excess time is also needed for quantum state preparation, associated entanglement operations, and measurement of expectation values within the hybrid interface. Looking at inference latency, the classical learner has an latency of 1.7 ms per sample, while the hybrid counterpart scaled the latency requirement to approximately 3.4 ms per sample. Although there is substantial increase in latency values, these models are practically viable for offline analytics and batch-guided deployment. In HPC computing environment, these hybrid models can be easily incorporated given their emphasis on the throughput efficiency (Table 4).

### 6.3.2 Algorithmic complexity analysis

To get a better understanding of the running performance, instances contributing in computing complexity of the proposed hybrid architectures should be properly evaluated. Table 5 illustrates the component-wise asymptotic complexity distribution of the proposed architecture. It is evident that the major classical computation cost is associated with the BILSTM backbone that required a complexity near to

$$\mathcal{O}(B \cdot T \cdot H^2), \tag{18}$$

in which $B$ is the batch size, $T$ is sequence length, and $H$ is the hidden dimension. This is majorly due to the floating-point operations required during training which can be easily handled by exploring efficient GPU acceleration and data parallelism. We have applied a global average pooling after the temporal encoding of data instance. This is responsible for converting sequence-level representation into a fixed-length semantic embedding. This operation required an average of $\mathcal{O}(B \cdot T \cdot H)$ cost. This pooling ensures that our further operations are not explicitly dependent on sequence length. After successful pooling, we used a dense layer of size $Q$ to project these embedding on a low-dimensional latent space. This overall operation needed a cost of $\mathcal{O}(B \cdot H)$ with a small constant $Q$. This step is the true classical-to-quantum interface. The quantum component introduced by us consists of a custom-featured VQC which is applied per sample. In general, for circuit build using $Q$ qubits with

**Table 4** Runtime and training cost comparison on 500k samples (mean ± std over 5 runs)

| Metric | Classical model | Quantum model |
|---|---|---|
| Epochs used | $7.8 \pm 1.8$ | $6.4 \pm 1.1$ |
| Time/epoch (s) | $\approx 647$ | $\approx 1170$ |
| Total train time (s) | $5044 \pm 1150$ | $7020 \pm 820$ |
| Per-step time (ms) | $\approx 275$ | $\approx 495$ |
| Inference latency (ms) | 1.70 | $\approx 3.4$ |

**Table 5** Algorithmic time complexity of major components in the proposed hybrid architecture

| Component | Complexity | Scaling factors |
|---|---|---|
| Token embedding | $\mathcal{O}(B \cdot T)$ | $B, T$ |
| BiLSTM encoder | $\mathcal{O}(B \cdot T \cdot H^2)$ | $H, T$ |
| Global average pooling | $\mathcal{O}(B \cdot T \cdot H)$ | $T, H$ |
| Quantum–classical projection (Dense) | $\mathcal{O}(B \cdot H)$ | $B, H$ |
| CF2 quantum projection | $\mathcal{O}(2^Q \cdot D)$ | $Q, D$ |
| Feature fusion | $\mathcal{O}(B \cdot H)$ | $B, H$ |
| Dense classifier | $\mathcal{O}(B \cdot H)$ | $B, H$ |

fixed depth $D$, the computational function for its classical simulation overhead can be formulated using the following:

$$\mathcal{O}(2^Q \cdot D). \tag{19}$$

In our proposed work, we have restricted our quantum circuit for a mere $Q = 5$ which have a state-space dimension of 32. It should be noted that our model operates on pooled representation and is used only once for an input sample. Thus, its computational cost is independent of the sequence length $T$. This way, we ensured that our quantum operations do not scale linearly with temporal depth which is very common in existing full recurrent quantum architectures. Finally, the feature fusion and dense classification layers contributed a runtime cost of $\mathcal{O}(B \cdot H)$. This cost is negligible to the overall BiLSTM backbone. Thus, it is evident that the computation instance is particularly dominated by the BiLSTM backbone in contrast with the quantum component which is executed within a bounded and sequence-independent overhead.

### 6.3.3 Scalability and HPC implications

We would like to notify the readers that reported quantum runtime within our study is executed over classically simulated VQCs. Thus, these measurement costs have to inferred as conservative upper bound for the quantum-enhanced training. If we further increase the qubits with denser entanglement structure or utilize multi-layer circuits, the simulation cost will overshadow the overall runtime cost exponentially.

Thus, to counter these shortcomings within current NISQ-associated architecture, the future research is needed to explore HPC options more profoundly. This will help in scalable quantum–classical learning with richer quantum components. For more expressive gate-level quantum recurrence, we need to explore robust accelerator-aware scheduling and parallel batch execution. The existing HPC infrastructure can enable a more prominent use of distributed simulation for heavier quantum circuits so that we can establish the true advantage of the current quantum enrichment. This way, we can easily balance the trade-off between representational capacity and computational cost of these approaches.

### 6.4 Ablation effectiveness of quantum layer

We carried out a systematic ablation study over three important quantum parameters: number of qubits, circuit depth, and data encoding strategy, in order to validate the contribution of the quantum layer and evaluate its sensitivity to architectural design decisions. This analysis aims to assess how quantum expressivity and computational cost trade-off within the suggested hybrid framework rather than to assert quantum advantage. We conducted the ablation experiments on a balanced subset of 100,000 samples from the Sentiment140 dataset (60% training, 20% validation, and 20% testing) to guarantee reproducibility and computational viability. This sample size was chosen to allow for the evaluation of several quantum configurations under the same conditions while maintaining the statistical features of the entire dataset. To isolate the effects of quantum design decisions, all classical elements, such as tokenization, embedding dimensions, BiLSTM architecture, optimizer settings, and training protocol, were kept constant throughout experiments. The ablation grid includes circuit depths 1,2, qubit counts 2,4,6, and encoding techniques angle, re-uploading. These ranges were selected to allow for significant variation in circuit expressivity while reflecting the practical limitations of near-term quantum simulation. While data re-uploading introduces repeated feature injection to increase nonlinear representational capacity without increasing qubit count, angle encoding represents a minimal baseline. By controlling the number of entangling operations and variational layers, circuit depth makes it possible to assess whether deeper circuits produce performance gains that are proportionate to their higher computational cost.

The resulting F1-scores and training times for each configuration are shown in Table 6. Based on these results, we can conclude various interpretation as follows. First, performance gains are only slight when the number of qubits is increased above a minimal configuration, suggesting diminishing returns in representational capacity for this task. Second, more entangling layers do not significantly improve sentiment discrimination in this hybrid setting, as shallow circuits (depth = 1) typically perform similarly to deeper circuits. Third, for moderate qubit counts, data re-uploading encoding marginally outperforms standard angle encoding. However, the observed performance gains are small relative to the additional computational overhead introduced by re-uploading. This supports our decision to adopt angle encoding in the proposed model, as it provides a more efficient trade-off between performance and complexity. Crucially, as qubit count and circuit depth increase,

there is increase in training time along with these monotonic performance gains. This proportional change reflects the trade-off between expressivity and computational overhead. Overall, the ablation results show that even at small scales, the quantum layer contributes significant nonlinear feature transformations; however, performance does not increase proportionately with excessive quantum complexity. This validates the design decision made in the primary experiments, which favors compact quantum circuits integrated as feature transformation modules over recurrent or deep quantum architectures.

### 6.5 Statistical significance analysis

We have conducted a paired $t$ test for statistically analyzing the F1-score that are obtained across five independent runs of both classical and quantum pipeline. This will enable us to introspect the true performance comparison between these models with identical data splits, preprocessing, and training. As hypothesized with the performance across various metrics described in previous section, the $t$ test supports no absolute performance gains across these models. We got a $t$-statistic of 0.0702 with a corresponding $p$ value of 0.9474 that establishes no statistical significance. The mean F1-score difference between these models $1.18 \times 10^{-4}$, with a 95% confidence interval of [−0.0045, 0.0048]. Cohen's $d$ was also evaluated to measure the effect size. Its value of 0.0314 underscores a negligible practical effect. From these results, we can conclude that although the difference and gain are negligible across aggregate evaluation metrics, the models still performed comparably. Thus, rather than contrasting our approach contribution for substantial performance gain, the architectural feasibility along with stable optimization should be a major starting point for parallel studies in future. Our proposed approach has the potential for advancing as an scalable architecture over large scale, though NISQ constraints needed to be considered and approached.

**Table 6** Quantum ablation study: effect of qubit count, circuit depth, and encoding strategy

| Qubits | Depth | Encoding | F1-score | Train time (s) |
|---|---|---|---|---|
| 2 | 1 | Angle | 0.7723 | 459.5 |
| 2 | 1 | Re-upload | 0.7738 | 481.5 |
| 2 | 2 | Angle | 0.7786 | 461.7 |
| 2 | 2 | Re-upload | 0.7792 | 504.4 |
| 4 | 1 | Angle | 0.7762 | 497.3 |
| 4 | 1 | Re-upload | 0.7780 | 534.5 |
| 4 | 2 | Angle | 0.7760 | 538.2 |
| 4 | 2 | Re-upload | 0.7800 | 613.4 |
| 6 | 1 | Angle | 0.7750 | 563.6 |
| 6 | 1 | Re-upload | 0.7805 | 623.4 |
| 6 | 2 | Angle | 0.7744 | 660.6 |
| 6 | 2 | Re-upload | 0.7751 | 782.0 |

## 7 Discussion

Our proposed hybrid quantum–classical learning pipeline displayed a comparable performance in comparison to classical pipeline. The proposed model achieves accuracy near to the classical learner in comparison with better consistency across runs. Although our approach does not produce substantial gains, its ability to consistently perform with the large-scale data is validated. Thus, future studies can consider our study as a valid reference to study more advanced quantum circuits that have richer feature representation methods, real hardware optimization, entanglement orientation, encoding strategies, and qubit structure. This will be essential for acquiring the real potential of quantum properties in the NLP domain.

### 7.1 Temporal sentiment dynamics of users: a case study

In addition to the aggregated performance metrics that quantified the overall sentiment classification accuracy of the proposed approach, we have also characterized this over time at the individual level. We used temporal weighted behavior metrics which are described in Sect. 4.2.3 to understand that how this approach can be integrated to monitor the user's emotional stigma in a real-time scenario. The following results help in structuring our efficacy in retrospecting user's longitudinal sentiment dynamics and population-level contrast.

#### 7.1.1 Case study user characterization

We selected a representative subset of four users based on temporal sentiment trajectories and emotional volatility statistics as described earlier. The details related to the selected users, their dominant sentiment trends, volatility levels, and associated behavioral regime are illustrated in Table 7. This provides a better contrast of the model's efficacy for social media-based mental health monitoring. In addition to these, Fig. 7 clearly visualizes these observed trends for these five users across the time period where distinct transition patterns are clearly visible.

#### 7.1.2 Recency-weighted sentiment dynamics

The affective evolution of selected 4 users based on their recency-weighted sentiment trajectory is shown in Fig. 7asmoothed and temporally contextualized affective mind state. We have transformed the posting history of users as time series and plotted on $x$-axis. For the $y$-axis we calculated the exponentially degraded aggregation probabilities as describe earlier. The positive $x$ values tell about the weighted positive state of mind in the user, while the negative aspect corresponds with the disturbed mind state. For the simplicity of analysis, we considered 0.0 as the neutral zone with no mental disturbance. This mental state threshold is just for descriptive purpose of the proposed architecture utility and should never be considered as an clinical-level mental health indicator. If it is to be used as an reliable indicator,

the threshold division needs to validate against the ground truth values under expert supervision in future studies.

We have visualized these user's state to represent the distinct longitudinal behavioral pattern across the dataset. The blue line representing *lonemat* showed strong negative emotional concerns initially, but with the passage of time, their mental state improved substantially. This user gradually recovered from extreme emotion conflict reaching near normalization zone. This can be attributed to past negative sentiment being attenuated with more recent reduced negative sentiments. In contrast, the orange line illustrates the emotional trajectory for *pawsthejaws*. This user experience frequent volatile swings around the neutral zone without any significant trend. This volatile emotional fluctuation across longitudinal aspect helps to demonstrate that these swings are emotional expressiveness rather than sentiment deterioration. Third user, `ICarluccio`, who is depicted by green line, has phased between both of the positive and negative sentiment regimes for the observed period. Initially, this user has positive affection supported near neutral zone. After some time, they transitioned toward negative zone due to prolonged deterioration in their mental health. In contrast to the *lonemat*, *hailey_letang* is showed using red line has experienced a reverse-directed (positive to negative) emotional transition over the period. These volatile transitions reflect how the tweet-level classification of mental state is inefficient in capturing these abrupt transitions. The user's may appear to be negatively influenced at the moment, but the core reason behind this state is not clearly understandable in per-data-based analysis. The long-term deterioration across mental sentiments can be only detected through structured longitudinal, recency-aware modeling.

#### 7.1.3 Emotional volatility patterns

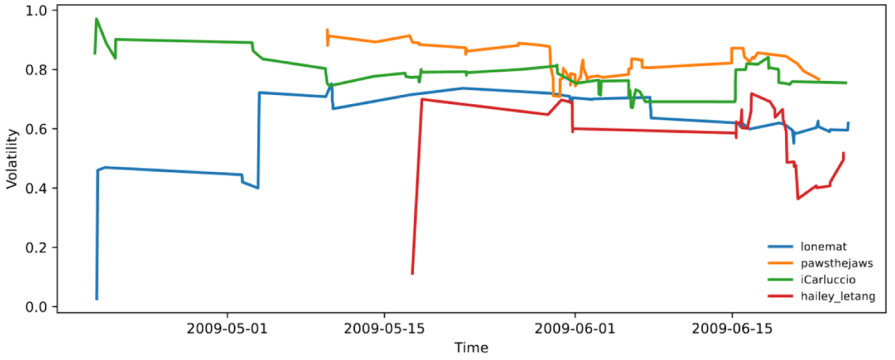
To get a more precise understanding of temporal evolution of emotional volatility, we have used rolling standard deviation of inferred sentiments over a time window. Figure 7b helps in understanding the emotional volatility aspect across selected users. The x-axis is plotted corresponding to the chronology of the user posts. The vertical *y*-axis plots the magnitude of volatility where higher values reflect more short-term emotional variability. It should be noted that this figure captures the degree of emotional reactivity over time in contrast to directive emotional trends in Fig. 7a. Based on these trends, it is evident that volatility is not necessarily having a proportional relation with sentiment polarity. If we consider both `pawsthejaws` and `ICarluccio`, they have shown consistent elevated volatility but experienced distinct long-term sentiment state. This underscores that emotional reactivity is dependent of frequent fluctuations even if their respective states are either near-neutral or transitions overtime. Notably, `lonemat` has comparatively lower volatility for majority of the time. This user experienced episodic emotional shifts rather than sustained instability. The red trajectory of `hailey_letang` showed moderate volatility, excluding the some initial timeperiod. The sudden volatility experienced by this user matches with sudden emotional deterioration observed in Fig. 7a

**Table 7** Temporal behavioral characterization of case study users

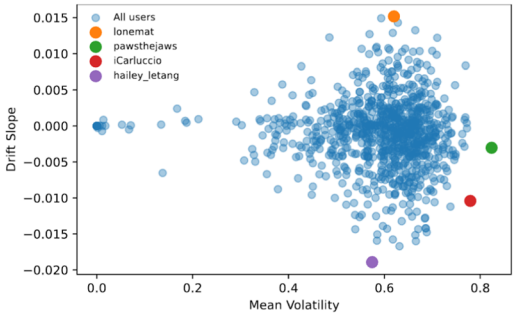
| User | Sentiment trend | Volatility level | Behavioral regime | Interpretation |
|---|---|---|---|---|
| lonemat | Negative-to-neutral stabilization | Moderate | Recovery/Stabilization | Initial negative affective expression followed by gradual stabilization toward neutrality, indicating recovery-oriented dynamics with episodic emotional variability |
| pawsthejaws | Near-neutral/oscillatory | High | Volatility-Dominated Expressive | Emotionally reactive behavior characterized by frequent short-term fluctuations without sustained affective drift, highlighting expressive variability rather than long-term sentiment change |
| ICarluccio | Multi-phase/near-neutral | High | Low Drift + High Volatility | Pronounced short-term emotional reactivity with identifiable transition phases, yet limited net sentiment drift over time, indicating non-stationary but balanced affective behavior |
| hailey_letang | Positive-to-negative drift | Moderate | Sentiment Deterioration | Sustained transition from positive affective expression to persistent negative sentiment, accompanied by increased variability during transitional periods, reflecting affective decline |




(a) Recency-weighted sentiment trajectories.




(b) Emotional volatility profiles over time.




(c) Drift-Volatility distribution of users.

**Fig. 7** Temporal case study and population-level comparison of user sentiment dynamics. **a** Recency-weighted sentiment trajectories demonstrate distinct longitudinal trends across users. **b** Emotional volatility profiles reveal expressive stability and instability independent of sentiment polarity. **c** Population-level drift–volatility distribution contextualizes the selected case study users within the broader dataset

The aforementioned use cases involving 4 users support the efficacy of our proposed hybrid quantum-enhanced learner to provide a multi-dimensional characterization of the mental health. We have jointly analyzed the recency-weighted sentiment trajectories, volatility patterns, and drift–volatility distributions at the

individual level. The recency-aware approach helps us in minimizing transient noise and preserves the meaningful temporal structure. So we established that snapshot-based methodology to assess the social media posts does not provide very intuitive and reliable clarity of users mental health. Thus, our approach enables to characterize the richer emotional dynamics of user.

## 8 Limitations and future directions

*a) Dataset scope and label semantics* Our current study has used only Sentiment140 dataset which has sentiment labels classified as positive or negative. Thus, the results obtained in the study through sentiment polarity should be interpreted as methodological proxy and not a direct indicator of psychology state. As this dataset lacks mental health-specific annotations, our approach might have failed to capture the clinical-specific temporal dynamics of user health. Our work evaluated the feasibility, stability, and scalability of the proposed architecture across large-scale data. Future studies looking in this aspect should specifically consider multi-class, regression labeling, and clinically grounded diverse datasets for more robust psychological state modeling and longitudinal assessment.

*b) Quantum simulation versus hardware execution* We have classically simulated the current quantum-enhanced circuit rather than using physical quantum hardware. We considered this option due to existing limitations associated with the NISQ hardware which are still limited with noise, qubit counts, and shorter coherence time. These real hardware circuits are also costlier to access as of now, and thus we validated our approach against real-world data through simulation approach. Thus, our reported results and runtime cost should be evaluated as conservative bound for computational requirement. Though simulation allowed us to perform our work under feasibly controlled stable training, future studies need to consider additional practical constraints. These studies should appropriately focus on gate noise, readout errors, and device-specific connectivity constraints. These factors directly influence the optimization and robustness of models. For more scalable experiments, the studies need to lookout for noise-aware training, cross-dataset evaluation, hardware-efficient circuit designs, quantum-enhanced attention, and feedforward block under practical operating conditions.

*c) Sensitivity analysis* We have evaluated the quantum-induced advantage of the hybrid framework using a predetermined set of design and hyperparameter selections. Although stable optimization and controlled comparison are made possible by this configuration, the model's sensitivity to individual components is not fully described. The trade-off between representational capacity and computational cost may be altered by changes in quantum circuit depth or qubit count, whereas the balance between short-term expressiveness and long-term trend preservation may be particularly impacted by changes in the temporal decay coefficient. Future studies should focus on more detailed component-wise ablation and sensitivity analyses to evaluate overall performance differences

between the classical and hybrid models. They need to compare configurations with and without quantum feature transformation under the same conditions, removing or altering the recency mechanism to isolate the effect of temporal weighting, and methodically changing the size and expressivity of the quantum layer. Such architectural and parameter ablations will give a better understanding of the relative contributions of each module and assist in identifying regimes where temporal aggregation or quantum-enhanced representations are most beneficial.

## 9 Practical implications

The proposed hybrid quantum–classical framework demonstrates the feasibility of integrating quantum-enhanced feature transformations with temporally aware sentiment modeling at scale. While the empirical evaluation focuses on sentiment-based affective analysis, the broader implications of such systems, particularly when applied to psychologically sensitive domains, require careful consideration of interpretability, ethical responsibility, and deployment constraints.

### 9.1 Scope and risk of misclassification

The resulting temporal signals are still indirect proxies of psychological state even though the framework uses temporally aggregated sentiment-derived indicators instead of isolated instance-level predictions. Social media expressions are inherently influenced by linguistic style, contextual framing, cultural norms, and situational factors. This introduces discrepancies between expressed sentiment and an individual's underlying mental condition. So, a false-positive indications may exaggerate emotional distress, while false negatives may obscure meaningful affective changes. These risks are particularly pronounced when sentiment polarity is interpreted without additional contextual or clinical information. The analysis presented in this study therefore treats inferred sentiment trajectories, volatility measures, and drift patterns as behavioral indicators of affective expression, rather than definitive assessments of mental health status. We have used longitudinal perspective that mitigates some of the instability associated with snapshot-based sentiment analysis by attenuating short-term expressive noise and emphasizing sustained trends. However, this temporal aggregation should be understood as a methodological enhancement rather than a substitute for clinically grounded validation. As such, the framework is best suited for exploratory monitoring and large-scale behavioral analysis, where the objective is to identify patterns and transitions over time rather than to draw individual-level diagnostic conclusions.

### 9.2 Human-interpretable temporal indicators

The temporal structure-based categories defined by us are intended to provide a clear understanding of study's practical perspective. In contrast to isolated tweet-level

predictions, these defined categories offer a higher-level perspective of user behavior by summarizing how emotional expression changes over time. These indicators help in describing patterns of variation and persistence in expressed sentiment rather than directly inferring psychological states. The used recency-weighted aggregation mechanism helps in reducing the influence of short-lived fluctuations while retaining sensitivity to gradual changes, enabling smoother trajectories that are more amenable to inspection. The degree of short-term variability is also captured by volatility measures, which do not make any assumptions about the direction of sentiment. Together, these indicators provide a descriptive description of long-term affective behavior that can be analyzed statistically or qualitatively. These temporal summaries are not intended to serve as conclusive markers of mental health or as diagnostic outputs. These indicators main function is only to organize and contextualize sentiment data over time, which could help with tasks like exploratory monitoring or downstream analysis. For a practical and valid interpretation of such signals, a strong knowledge and external context is necessary.

### 9.3 Deployment considerations

The proposed framework is designed to operate as a modular analytics component that can be integrated into real-time public health surveillance pipelines or decision support dashboards, rather than functioning as a standalone diagnostic system. The framework is most appropriate for large-scale exploratory monitoring and decision support based on social media data, rather than for autonomous or fully automated psychological assessment. However, we would like to draw the attention to the fact that the temporal indicators derived in this study should be understood as contextual signals that assist analysts, clinicians, or public health experts in identifying evolving trends or typical patterns that may warrant further attention. Any realistic system deployment must incorporate human-in-the-loop integration to address this. This will help in reducing the associated risks of misclassification, overinterpretation, and contextual misunderstanding while ensuring that algorithmic outputs do not take precedence over expert judgment. Our framework's emphasis on aggregated longitudinal behavior rather than instantaneous predictions through temporal analysis further supports this supporting role. In clinical or organizational settings, the framework can therefore support longitudinal visualization of affective trends for expert interpretation and population-level monitoring through plug-in real-time monitoring.

## 10 Conclusion

In this work, we evaluated the social media posts for getting a scalable and realistic understanding of user's mental state. Our proposed hybrid quantum-enhanced learner integrated BiLSTM architecture with the custom-featured quantum circuit. The classification probabilities are then aggregated to characterize the temporal sentiment distribution of the users. Our proposed architecture achieved performance

comparable to the classical baseline on the twitter dataset. This hybrid quantum classifier converges much faster than its counterpart with similar accuracy, precision, recall, and F1-score. These results underscore the efficacy of introducing quantum learner within classical architectures without loosing the performance capability. We have further proposed an integration of temporal sentiment modeling with classification pipeline. The suggested framework reveals longitudinal user-level dynamics that are not visible through static sentiment predictions alone by fusing emotional volatility estimation with recency-weighted sentiment aggregation. We have structurally classified the user's behavioral pattern through drift and volatility trajectories. Overall, this work demonstrates the usefulness of hybrid quantum–classical recurrent architectures in temporally aware sentiment analysis and establishes their viability and resilience for large-scale NLP tasks. Our results provide a reproducible foundation for future research on deeper quantum circuits, richer temporal representations, and clinically grounded datasets as quantum hardware and simulation capabilities continue to advance.

**Author contributions** G.P. carried out data preprocessing, model implementation, experiments and prepared the initial draft of the manuscript. K.S.R. provided supervision, guidance on methodology, critical revisions, and overall manuscript editing. T.S. made substantial contributions by addressing reviewer concerns, updating analyses, and revising the manuscript contents. All authors reviewed and approved the final version of the manuscript.

**Data availability** To support reproducibility and transparency, all code associated with this study is publicly available at:https://github.com/wakai-megumi/Mental-Health-aware-Sentiment-Analysis-using-Quantum-Features.

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## Declarations

**Conflict of interest** The authors declare no conflict of interest.

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